

ATMOSPHERIC TURBULENCE MEASURED IN STARS ABOVE THE MAIN SEQUENCE

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ABSTRACT

Measurements of broadened line profiles are analyzed to identify, measure, and separate macroturbulence from rotation. Evidence is presented for the possibility of more than one macroturbulence velocity distribution.

Subject headings: line profiles — rotation, stellar — turbulence

I. BACKGROUND

Rotation and macroturbulence have traditionally been regarded as spectral line broadening mechanisms, but several investigators have shown how difficult it is to tell which of these mechanisms is responsible for the observed broadening (Slettebak 1956; Abt 1958; Kraft 1969, 1970). Some stars show lines so broadened that velocities of hundreds of km s^{-1} are indicated. Such numbers are possible with rotation but implausibly high for turbulence. Studies of eclipsing binaries especially have given strong evidence for stellar rotation (Rossiter 1924; McLaughlin 1924; Shajn and Struve 1929). Evidence for macroturbulence is much less compelling. It was first suggested as a line broadening mechanism by Evershed (1922). Macroturbulence certainly exists in the Sun with characteristic velocities of $1\text{--}2 \text{ km s}^{-1}$. Macroturbulence has been invoked in stellar investigations for example by Huang and Struve (1953), Groth (1961), Wolf (1971), Scholz (1972), and others. In all of these analyses, an isotropic Gaussian velocity distribution is *assumed* for the macroturbulent elements.

The purpose of this paper is to show explicitly how macroturbulence can be distinguished and separated from rotation and how information about the macroturbulence velocity distribution can be obtained. Several examples are given to illustrate the procedure which is an application of the technique proposed earlier (Gray 1973, Paper A).

II. DATA

Line profiles of the $\text{Mg II } \lambda 4481$ doublet, $\text{Fe II } \lambda 4508$, or $\text{Fe I } \lambda 6065$ and $\text{Ti I } \lambda 6064$ were measured using the photoelectric line scanner described by Gray (1973) and by Gray and Evans (1973). The instrumental profile was carefully measured using a ^{198}Hg lamp. Half-widths of $0.20 \text{ \AA}$ and less were used for all the measurements included here, with the individual instrumental profiles being measured separately for each set of data. The stellar profiles were scanned several times, being sampled in equal wavelength steps. The size of the wavelength step, and consequently the Nyquist frequency in the Fourier transforms, is dif-

ferent for different lines because the spectrograph dispersion is a function of wavelength and because a smaller step in millimeters was adopted for the narrower lines. The dark-corrected photon counts were summed for each position in the profile, ratioed by the reference channel counts, and normalized to unit continuum flux. Random photometric errors of 1-3 percent of the continuum are typical for these data. (The data for $\alpha \text{ Peg}$ are the poorest presented here.)

The profiles were assumed to be symmetric, no evidence being found to the contrary, and the two sides of the profile were averaged in order to increase the signal-to-noise ratio. But for only two of the profiles ($\gamma \text{ Cyg}$ and $\beta \text{ Ori}$) did the data sampling happen to occur in an almost perfectly symmetrical way. In these two cases, the pairs of profile ordinates were simply averaged, but more generally it was necessary to draw a line through the data points as illustrated in Figure 1. The averaged values were read back from the continuous profile at the wavelength spacing of the original data. Any artificial smoothing resulting from this process will appear toward the highest Fourier frequencies so that it does not affect the analysis which,

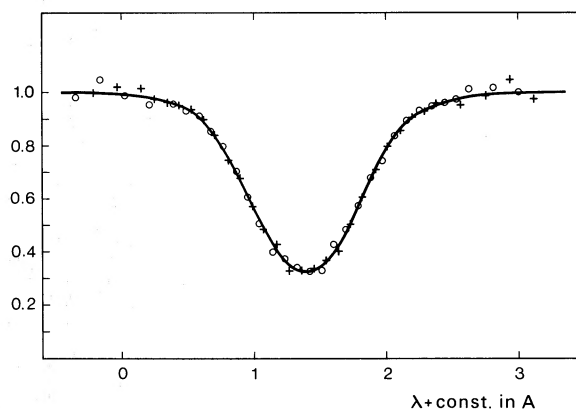


FIG. 1.—Observed $\lambda 4481$ profile for $\alpha \text{ Cyg}$. Circles, actual measurements; crosses, same points reversed in λ and superposed. The smooth curve is drawn in by hand.

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TABLE 1
THE MEASURED STARS

Star (1)	HR (2)	Sp (3)	Date of Observation (4)	λ (Å) (5)	$v \sin i$ (catalog) (6)	$v \sin i$ (7)	ζ^* (8)	ξ (9)
β Ori A.....	1713	B8 Ia	1973 Dec. 16/17	4481	42	42	23 RT	...
α Peg.....	8181	B9.5 III	1974 Nov. 29/30	4481	149	132	...	...
α Cyg.....	7924	A2 Ia	1973 Dec. 17/18	4481	21	...	42 RT	8.5
γ Cyg.....	7796	F8 Ib	1973 Oct. 15/16	4508	20	...	14.5 G	5.3
β Oph.....	6603	K2 III	1973 July 18/19	6065	< 17	...	5.9 RT	1.8
				6064				
γ^1 And.....	603	K3 II	1973 Sep. 3/4	6065	< 17	...	5.1 G	2.0
				6064				
γ Aql.....	7525	K3 II	1973 Sep. 12/13	6065	< 17	...	7-8 RT	2.5
				6064				

* The letters following the ζ value indicate a Gaussian or radial-tangential velocity distribution.

as we shall see, is done at frequencies well below the Nyquist frequency. It was also necessary to depend on the continuous profile to correct for the weak blend on the long-wavelength shoulder of $\lambda 6065$.

The seven stars listed in Table 1 were observed on the dates shown in column (4). They were selected to illustrate the technique. Their only other virtue in this context is that they are apparently bright. Table 1 also

TABLE 2
THE AVERAGED LINE PROFILES

Star.....	β Ori	α Peg	α Cyg	γ Cyg	β Oph		γ^1 And		γ Aql	
Line.....	$\lambda 4481$	$\lambda 4481$	$\lambda 4481$	$\lambda 4508$	$\lambda 6065$	$\lambda 6064$	$\lambda 6065$	$\lambda 6064$	$\lambda 6065$	$\lambda 6064$
$\Delta\lambda$ (Å).....	0.0924	0.231	0.0924	0.0461	0.0416	0.0416	0.0416	0.0416	0.0416	0.0416
	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000
	995	999	1025	990	997	996	996	998	999	999
	1000	992	1000	1010	994	991	993	995	993	992
	979	977	1003	985	992	982	990	988	986	980
	978	953	985	984	987	978	989	980	978	968
	979	934	970	986	978	940	982	958	962	930
	957	910	968	972	960	890	968	908	942	875
	904	898	955	936	930	793	942	823	910	798
	850	886	934	910	875	700	913	738	855	680
	806	879	910	854	763	610	862	628	760	545
	753	875	858	784	645	583	795	542	635	472
	712	875	790	720	514	653	680	520	525	462
	638	877	705	642	377	750	563	555	410	503
	611	880	610	530	310	845	435	655	306	590
	584	883	513	435	330	916	336	760	270	720
	589	890	425	343	420	950	288	845	272	825
	584	902	370	272	560	978	295	917	330	890
	611	920	335	235	680	985	375	966	446	942
	638	935	335	213	810	990	475	988	560	977
	712	960	345	235	890	994	600	996	670	990
	753	980	412	272	932	998	730	1000	760	997
	806	990	483	343	958	1000	826	...	850	1000
	850	996	573	435	970	...	890	...	911	...
	904	999	678	530	980	...	936	...	950	...
	957	1000	753	642	983	...	958	...	969	...
	979	...	836	720	988	...	974	...	980	...
	978	...	892	784	990	...	984	...	988	...
	979	...	928	854	993	...	990	...	992	...
	1000	...	947	910	996	...	994	...	998	...
	995	...	958	936	1000	...	998	...	1000	...
	1000	...	965	972	...	...	1000	...	...	...
	...	...	986	986	...	...	...	...	...	...
	...	...	988	984	...	...	...	...	...	...
	...	...	1014	985	...	...	...	...	...	...
	...	...	1000	1010	...	...	...	...	...	...
	...	...	...	990	...	...	...	...	...	...
	...	...	...	1000	...	...	...	...	...	...

lists the HR numbers and the spectral types from the Bright Star Catalog (Hoffleit 1964) and the $v \sin i$ values (col. [6]) according to the catalog of Uesugi and Fukuda (1970). The remaining numbers in Table 1 are a summary of the results obtained below.

Table 2 gives the profiles after averaging of the two sides of the profile. The wavelength step is given at the top of each column.

III. METHOD OF ANALYSIS

In earlier work (Paper A) the Fourier analysis procedure for obtaining information from line profiles was outlined. The part of the method emphasized here hinges on two relations. The first is for rotation. The line profile in the flux spectrum is

$$\mathcal{F}_v = G(\lambda) * \mathcal{F}_v^0, \quad (1)$$

where the flux profile for the nonrotating star is $\mathcal{F}_v^0$, the asterisk denotes convolution, and $G(\lambda)$ is the rotation profile. For our purposes the usual analytical expression for $G(\lambda)$ is adequate.

The second equation upon which the present discussion is based is for the profile broadened by macro-turbulence,

$$\mathcal{F}_v = \oint I_v * \Theta(\lambda) \cos \theta d\omega, \quad (2)$$

where $\Theta(\lambda)$ is the macroscopic velocity distribution projected onto the line of sight and I_v and $d\omega$ are specific intensity and solid angle increment as in the usual flux integral.

If $\Theta(\lambda)$ is assumed to be isotropic and therefore independent of the angular coordinate θ , then equation (2) becomes

$$\begin{aligned} \mathcal{F}_v &= \Theta(\lambda) * \oint I_v \cos \theta d\omega \\ &= \Theta(\lambda) * \mathcal{F}_v^0, \end{aligned} \quad (3)$$

where $\mathcal{F}_v^0$ denotes the intrinsic flux profile. We make no assumption about the form of this isotropic $\Theta(\lambda)$ but instead derive it as part of the analysis.

A second model for the macro motions is the non-isotropic case where a fraction A_R of the emitting area is moving along stellar radii and the remaining fraction, A_T , is moving tangentially. Then by assuming that I_v in equation (2) is independent of θ and the distribution of velocities in the radial directions and in the tangential directions are Gaussian, one finds (see Gray 1975 for greater detail)

$$\begin{aligned} \mathcal{F}_v &= 2\pi I_v * [A_R Q(\zeta_R) + A_T Q(\zeta_T)] \\ &= \mathcal{F}_v^0 * 2[A_R Q(\zeta_R) + A_T Q(\zeta_T)], \end{aligned} \quad (4)$$

and

$$Q(\zeta) = (\pi\zeta)^{-1/2} (\lambda/\zeta) \int_0^{\zeta/\lambda} \exp(-1/u^2) du.$$

The Gaussian dispersion is ζ , and λ is the wavelength measured from the line center. In our use of equation (4), we take either $A_R = 1$, $A_T = 0$, or equivalently $A_T = 1$, $A_R = 0$, or $\zeta_R = \zeta_T$ (all three cases produce the same effect). Incidentally, the neglect of limb darkening will affect the radial and tangential contributions in the opposite sense. This amounts to changing the areal weights A_R and A_T .

The model expressed in equation (4) has the same form as the isotropic models of equation (3), namely, a macroturbulence broadening function convolved with the intrinsic flux profile $\mathcal{F}_v^0$.

The data, $D(\lambda)$, with which we deal will be a combination of the intrinsic flux profile, rotation, macroturbulence, and the instrumental profile of the spectrograph, denoted by $I(\lambda)$. We write

$$D(\lambda) = I(\lambda) * M(\lambda) * G(\lambda) * \mathcal{F}_v^0, \quad (5)$$

where $M(\lambda)$ is $\Theta(\lambda)$ if equation (3) is used, or the corresponding terms in equation (4) if it is used. Each of the functions $I(\lambda)$, $M(\lambda)$, and $G(\lambda)$ is normalized to unit area, so the equivalent width is the same for $D(\lambda)$ and $\mathcal{F}_v^0$. In the Fourier transform domain the convolutions become products, and we write

$$d(\sigma) = i(\sigma)m(\sigma)g(\sigma)f(\sigma), \quad (6)$$

where the lowercase symbols denote the transforms of the functions in equation (5) and σ is the Fourier frequency. We can obtain any of the transforms on the right by dividing equation (6) by the other transforms.

The procedure then is to compute $d(\sigma)$ from the observed profile, remove the instrumental profile by division with $i(\sigma)$, remove the thermal profile by division with $f(\sigma)$, and interpret the residual transform in terms of $g(\sigma)$ or $m(\sigma)$ or their product. This amounts to a dissection of the line profile according to the physical mechanisms assumed to be at work in broadening the observed line. It is the opposite of the usual process of fitting a line profile where the profiles of the various physical mechanisms are synthesized together and then compared to the data.

IV. EXAMPLES OF APPLICATION: ROTATION

Consider first the case of Mg II $\lambda 4481$ in α Peg as shown in Figure 2. The data transform, $d(\sigma)$, after division by $i(\sigma)$ is denoted by squares. The effect of the instrumental profile is so small in this case that the data points prior to division by $i(\sigma)$ are within the size of the squares and are omitted from the figure. The $\mathcal{F}_v^0$ profile is computed with an LTE model atmosphere of the type described in Paper A, with a solar Mg abundance, and with a classical microturbulence dispersion ξ chosen to make the equivalent width equal to the observed value.¹ Upon removal of the $\mathcal{F}_v^0$ profile, each square rises to the triangle above it at the same σ to give a point on the residual transform.

¹ This approximate way of establishing $\mathcal{F}_v^0$ is adequate for this star. The solution is nearly independent of $\mathcal{F}_v^0$ because the rotation dominates the observed profile. A more rigorous approach is used in the next section.

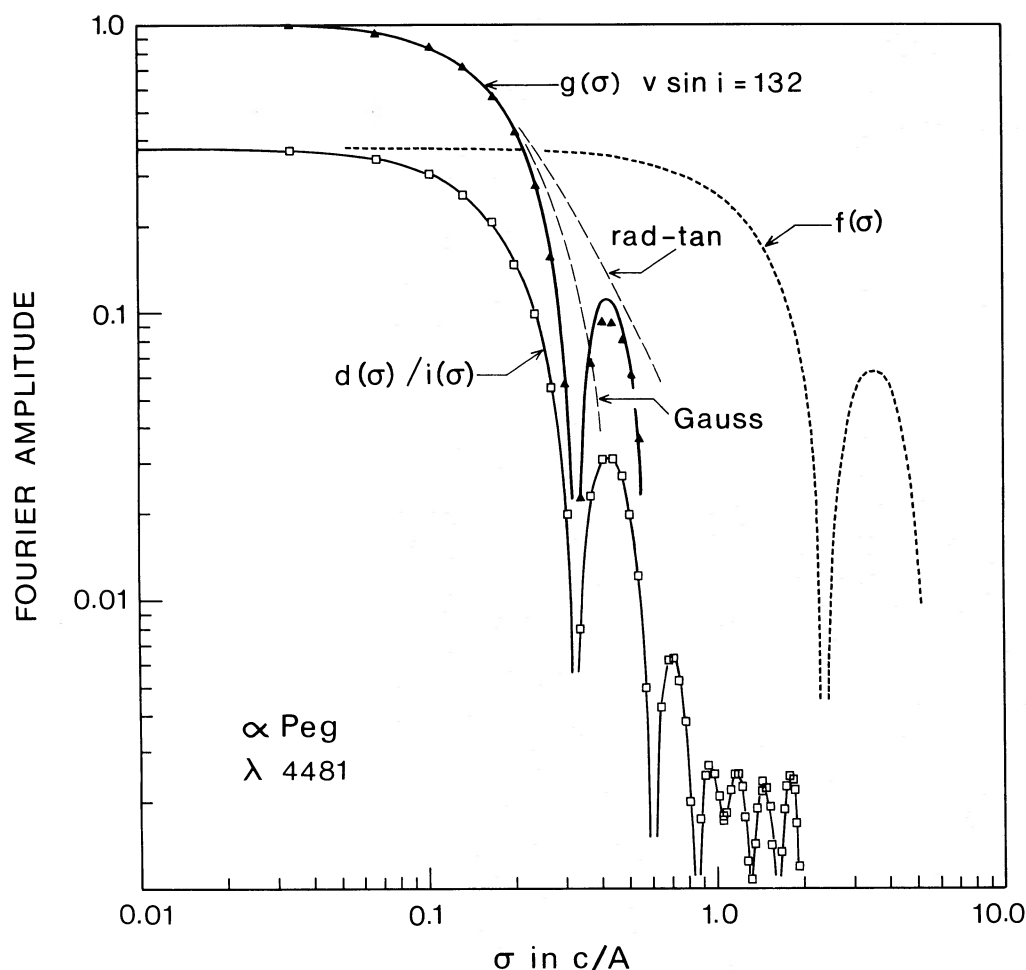


FIG. 2.—The dissection analysis for $\lambda 4481$ in α Peg. The residual transform (*triangles*) is indicative of rotation but not macro-turbulence.

We see that a very characteristic shape has emerged—the shape of a rotation profile transform, $g(\sigma)$, including the first sidelobe. Two macro-turbulence transforms are also shown in Figure 2 from which it can be seen that macro-turbulence is unlikely to be the cause of the broadening.

Plotting $g(\sigma)$ for $v \sin i = 1 \text{ km s}^{-1}$ on the same log-log coordinates allows us to find the star's $v \sin i$ by translating the $g(\sigma)$ plot along the abscissa until it matches the residual transform. A comparison of abscissas gives $v \sin i$. A $v \sin i$ of 132 km s^{-1} is found for α Peg. There is no measurable macro-turbulence in this star since $g(\sigma)$ accounts completely for the residual transform within the errors of measurement. The error associated with each point in any of these transforms increases with σ as the transform amplitudes diminish toward the noise level. It follows that we must require the curves to fit the low σ points more precisely than the higher σ points. The small differences in amplitude between the theoretical and the observed sidelobe is within the uncertainty of these measurements. Additional comments about errors are made later.

The removal of the instrumental and the thermal plus microturbulence profiles has only a small effect on the residual transform. The filtering by $i(\sigma)$ and $f(\sigma)$ is small in this case because $v \sin i$ is large. For stars showing smaller $v \sin i$, the filtering becomes proportionately larger. Eventually, for $v \sin i$ small enough, the first sidelobe will be reduced to the noise level. Then it will be more difficult to unambiguously identify the broadening as rotation because the interpretation will depend on the characteristic shape of the main lobe alone. It follows from these comments that the intrinsic profile, $\mathcal{F}_v^0$, does not matter much for stars having large $v \sin i$. Uncertainty in the microturbulence, for example, introduces a negligible uncertainty in the result.

V. EXAMPLES: K GIANTS

Let us now turn to the three K stars, β Oph, γ^1 And, and γ Aql. A comparison of the data transforms for the saturated Fe I $\lambda 6065$ line with those for the weaker

Ti I $\lambda 6064$ line² shows in each case that the first zero always occurs at a smaller σ for $\lambda 6065$. If the profile shapes were dominated by rotation or macroturbulence (as was the case for α Peg), then the first zero would arise from the first zero in $g(\sigma)$ or $m(\sigma)$ and would be at the same σ for both lines. Since this is not the case for these K stars, we conclude that the first zero is in $f(\sigma)$ with its position dependent on the degree of saturation of the line.

Profiles, $\mathcal{F}_v^0$, for $\lambda 6065$ were then calculated using model atmospheres with temperatures chosen according to the $B - V$ color index of the star. It is not necessary to know the temperature accurately. The transforms of these profiles were required to have the following characteristics: (1) The first zero is at or near the position of the first zero in the data, $d(\sigma)/i(\sigma)$. This amounts to matching the widths of the observed and computed profiles since the width of any function and its transform are inversely related. (2) The ordinate at $\sigma = 0$, which is the equivalent width of the line, must be the same for $f(\sigma)$ and $d(\sigma)$. (3) The transform $f(\sigma)$ must always be greater than or equal to $d(\sigma)/i(\sigma)$ in order that the removal of $f(\sigma)$ not produce a residual transform in excess of unity.

The reason why unity is the maximum allowed value for the residual transforms can be easily seen by reflecting on the definition of the transform of a real and positive function such as $M(\lambda)$; namely,

$$\begin{aligned} m(\sigma) &= \int_{-\infty}^{\infty} M(\lambda) e^{2\pi i \lambda \sigma} d\lambda \\ &= \int_{-\infty}^{\infty} M(\lambda) \cos(2\pi \lambda \sigma) d\lambda \\ &\quad + i \int_{-\infty}^{\infty} M(\lambda) \sin(2\pi \lambda \sigma) d\lambda. \end{aligned}$$

Since the cosine and sine have some negative values for $\sigma \neq 0$, and $M(\lambda)$ is positive, the integrals will have smaller values for $\sigma \neq 0$ than for $\sigma = 0$. The function $M(\lambda)$ is normalized to unit area so the $\sigma = 0$ value of $m(\sigma)$ is unity, the maximum value. The function $m(\sigma)$ can be thought of as a filter which reduces the amplitudes of $f(\sigma)$ to $m(\sigma)f(\sigma)$ in accord with equation (6). The same could have been said using $G(\lambda)$ in place of $M(\lambda)$.

Condition number 3 was used in Paper A to show that classical isotropic, Gaussian, depth-independent microturbulence can be used to remove the unphysical residuals (those greater than unity) which occurred when no microturbulence was allowed. The same situation exists in these stars but even more strongly. In fact, the three conditions are so restrictive that only one solution could be found; i.e., only one pair of the variables iron abundance and microturbulence dispersion, ξ , which are the variables controlling the line

² The Fe I line has a wavelength of 6065.49 Å; the Ti I line, 6064.63 Å.

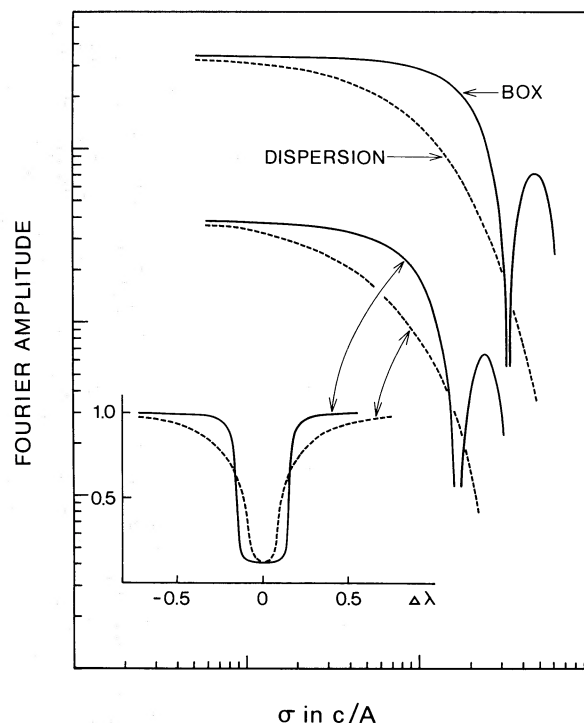


FIG. 3.—Top, the transform of a dispersion profile and a box. The characteristics of these functions are seen in the transforms of the two profiles shown in the insert graph. In particular, the line with strong wings has a transform showing a pointed rather than a flat top on the main lobe.

strength, will meet the three conditions. In each case the abundance was approximately solar (within a factor of 2).

Since the deduced form of the macroturbulence velocity distributions as well as some other conclusions depend upon these points, let us elaborate slightly. A profile with no microturbulence broadening can reach the measured equivalent widths of these strong lines ($\lambda 6065$) only by having substantial wings. The presence of wings introduces into the line's transform the characteristics of the dispersion function transform as shown in Figure 3. One characteristic of such a transform is a pointed main lobe. A profile of a saturated line having microturbulence is much boxier, making the main lobe of the transform very flat. Now the main lobes in the transforms of the observed profiles are flat. We see immediately that the pointed theoretical transforms cannot be used with the flat-topped data transforms and still satisfy the three conditions stated above. Strong wings are therefore excluded from $\mathcal{F}_v^0$; and, with the models being used, microturbulence is the only variable left by which the large equivalent width can be obtained. In addition, microturbulence which is depth-independent or increasing outward produces the needed boxiness in $\mathcal{F}_v^0$.

[The reader may wonder why not instead simply look for wings or their absence in the profile directly. If there were no other broadening mechanisms, such a

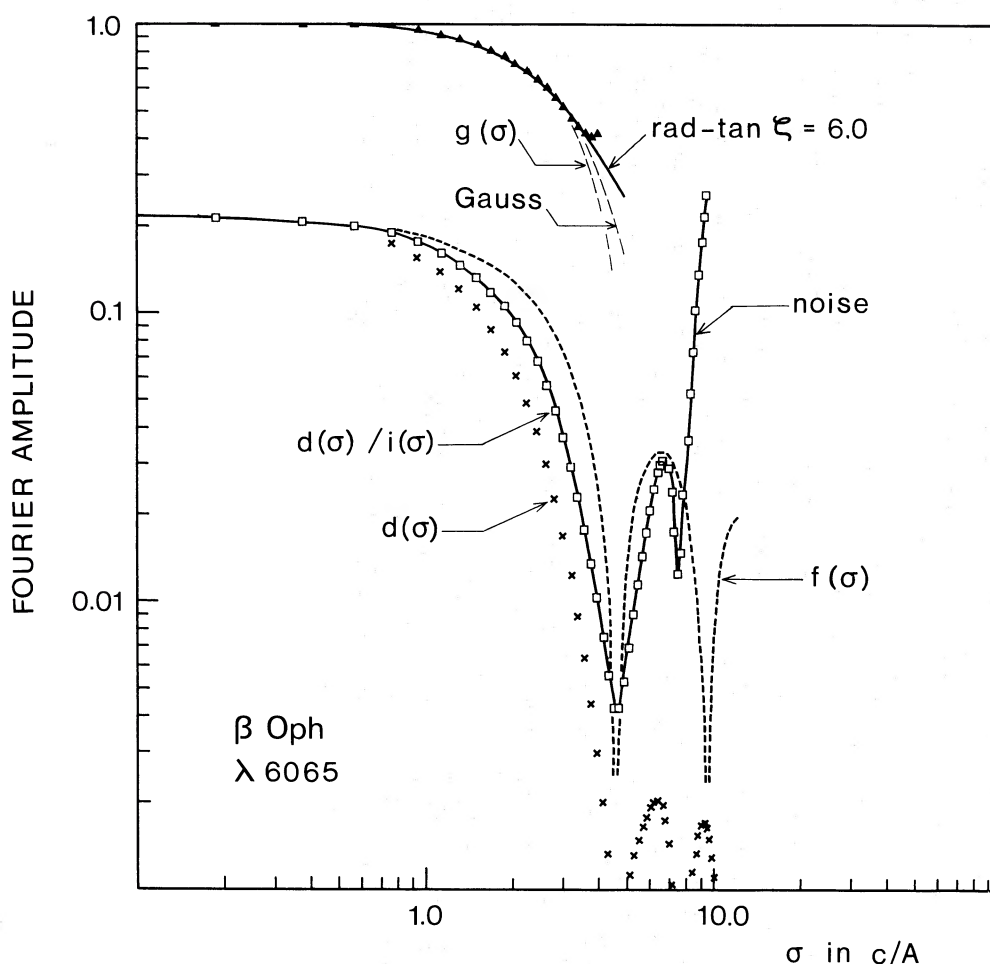


FIG. 4.—The analysis of $\lambda 6065$ in β Oph. The thermal plus microturbulence ($\xi = 1.8$) profile is chosen to satisfy the three conditions stated in the text. The residual transform is shown by triangles. Macroturbulence fits better than rotation. The squares beyond $\sigma \sim 4\frac{1}{2} c/\text{\AA}$ are noise.

scheme would work. Of course, here we explicitly have other broadening mechanisms such as the instrumental profile that will produce a smearing of the profile and give it wings. In order to inspect the profile directly, then, we must first remove the instrumental profile. This means more work in computing the inverse transform and, more importantly, it means choosing a noise filter to eliminate the magnified high frequency noise caused by the division of $d(\sigma)$ with $i(\sigma)$. See Brault and White (1971) or Gray (1975). But all of this is avoided by working directly in the Fourier domain.]

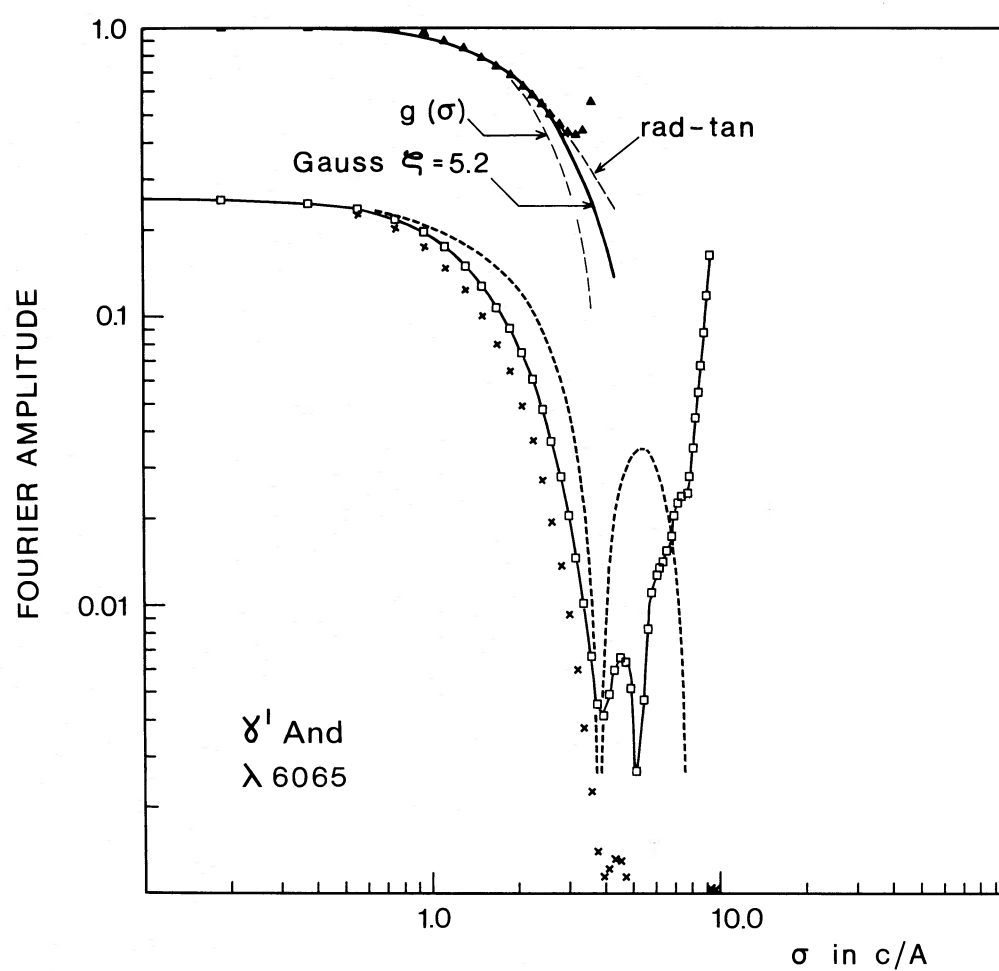
The plan then for the three K stars at hand is to first analyze the saturated $\lambda 6065$ line from which we find the microturbulence dispersion ξ and the lowest Fourier frequency components of the residual transforms. Then we go on to the weaker $\lambda 6064$ line, and using the ξ from $\lambda 6065$, explore the residual transforms to somewhat higher Fourier frequency.

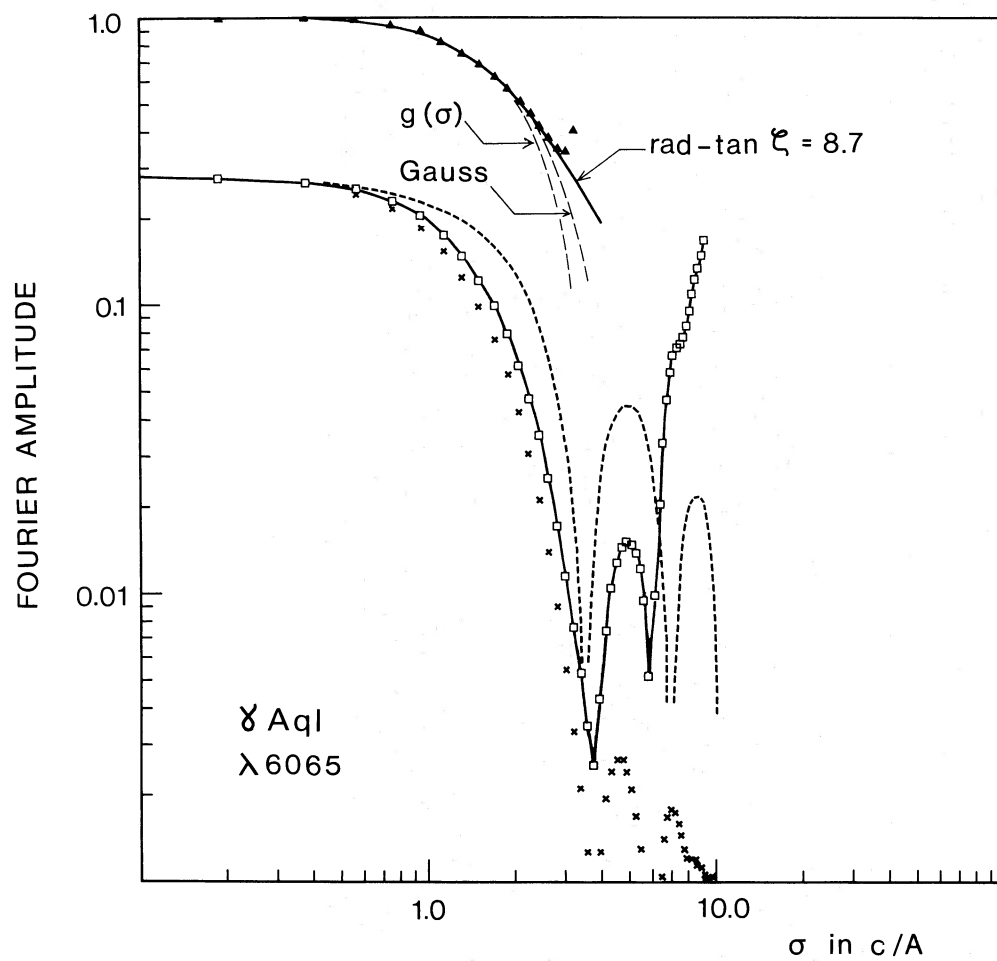
Figures 4, 5, and 6 show the dissection analysis for $\lambda 6065$. The $f(\sigma)$'s have been determined according

to the three criteria explained above. The residual transforms are compared to three interpretive functions corresponding to equations (1), (3) with a Gaussian $\Theta(\lambda)$, and (4). Rotation can probably be ruled out in all three cases. The choice between the macroturbulence models is not clearly defined, particularly in view of the larger errors associated with the last few points on the high- σ end of the residual transforms.

The reason why the residual transforms for $\lambda 6065$ do not extend to higher σ (and allow a choice between the macroturbulent models) is because of the filtering of $f(\sigma)$. The filtering for the weaker Ti I $\lambda 6064$ line is less, however, and we proceed on to these observations for additional information.

Figures 7, 8, and 9 show the $\lambda 6064$ analyses. We use the same model atmospheres including the same microturbulent ξ values that were used for $\lambda 6065$ and simply adjust the Ti abundance to obtain the observed equivalent width. The residual transforms for $\lambda 6064$

FIG. 5.—The analysis of $\lambda 6065$ in $\gamma^1 \text{ And}$

FIG. 6.—The analysis of $\lambda 6065$ in γ Aql

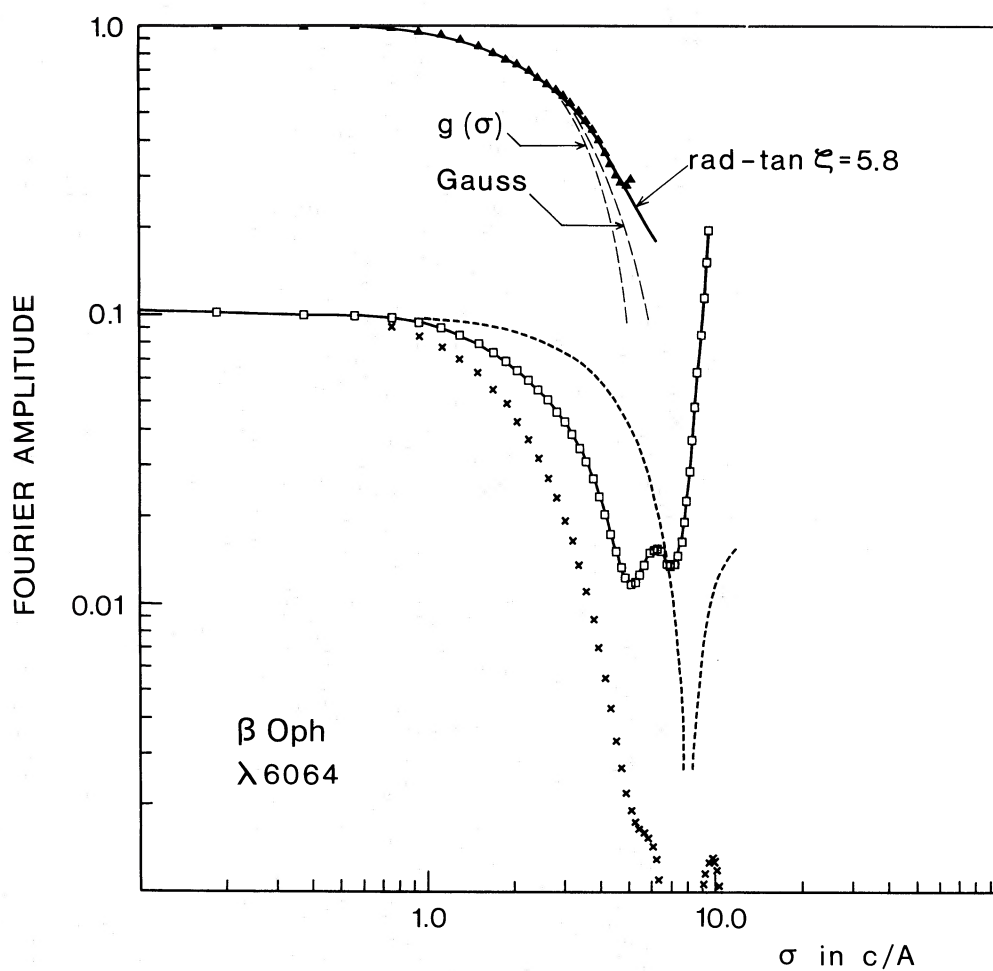


FIG. 7.—The analysis of $\lambda 6064$ in β Oph. The thermal plus microturbulence ($\xi = 1.8$) profile is a weaker filter than it was for $\lambda 6065$ and the residual transform can be traced to a higher σ . A radial-tangential macroturbulence with $\xi = 5.8 \text{ km s}^{-1}$ is indicated. Again, the noise becomes dominant for $\sigma \gtrsim 5c/\Delta$.

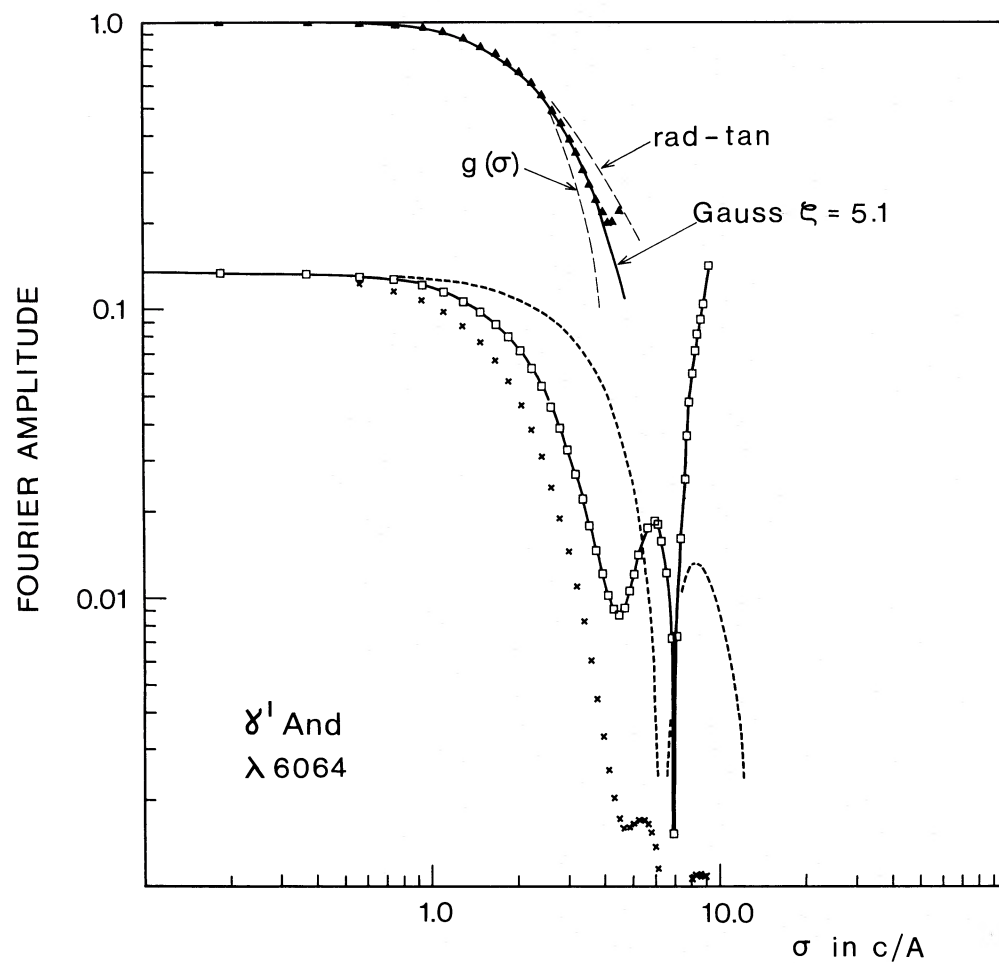


FIG. 8.—The analysis of $\lambda 6064$ in γ^1 And. A Gaussian macroturbulence with $\xi = 5.1 \text{ km s}^{-1}$ is indicated.

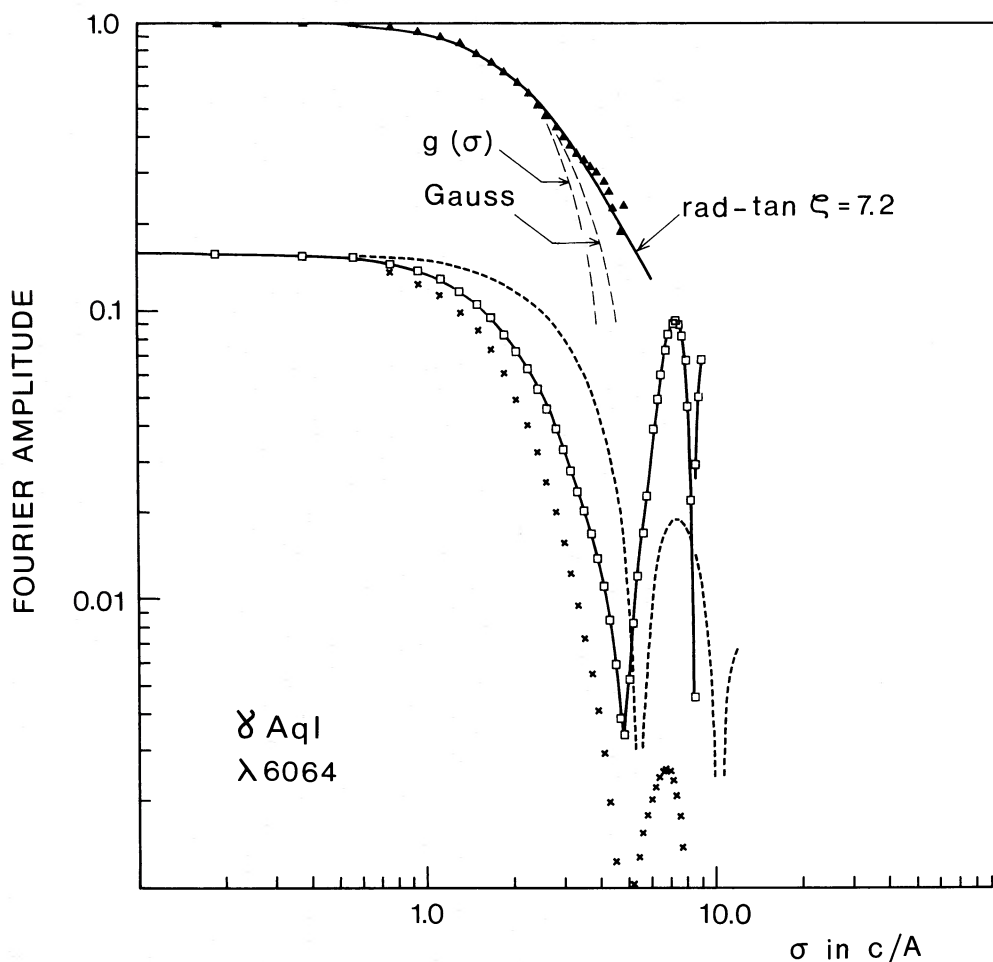


FIG. 9.—The analysis of $\lambda 6064$ in γ Aql. A radial-tangential macroturbulence with $\xi = 7.2 \text{ km s}^{-1}$ is indicated.

do extend to high enough σ to give information about $m(\sigma)$ and hence $\Theta(\lambda)$. Rotation is now clearly ruled out. We see that β Oph and γ Aql show radial-tangential (eq. [4]) behavior while γ^1 And shows Gaussian (eq. [3]) behavior. The values of dispersions are tabulated in the last columns of Table 1.

The values of the macrovelocity dispersions indicated by $\lambda 6065$ and independently by $\lambda 6064$ agree to about $0.1\text{--}0.2 \text{ km s}^{-1}$ for two of the stars. The third shows 1.5 km s^{-1} difference, which may be more indicative of the true uncertainty.

VI. EXAMPLES: SUPERGIANTS

The Fe II $\lambda 4508$ line of γ Cyg has been measured. The observed central depth of the line is nearly 80 percent, and the equivalent width is about $450 \text{ m\AA}$. Such a line cannot be broadened mainly by macroturbulence or rotation, but must have a substantial microturbulence unless the wings are very large. A glance at the line's transform in Figure 10 convinces us that the line wings must be small, because the main

lobe has a flat top. We are led to try the analysis used on the K giants. A solution can almost be obtained, but residuals slightly larger than unity are found even in the closest solution. It is possible that the wayward residuals (which are much smaller than those dealt with in Paper A) can be attributed to observational errors. On the other hand, two alterations in the model are possible which make the analysis flow smoothly again. One is to allow the microturbulence dispersion, ξ , to increase outward through the atmosphere. The other is to allow a lower boundary temperature than that given by the usual models. The second alternative was adopted because it is likely that larger line blanketing exists in supergiants compared to main-sequence stars and because the non-LTE effects will lower the source function toward the outer boundary—an effect which is simulated in an LTE model by dropping the temperatures toward the outer boundary. The consequence is a deeper line core in the model lines, allowing a few percent increase in the equivalent width for a given half-width. The conclusions of the analysis are not significantly altered if the boundary temperature adjustment is not made.

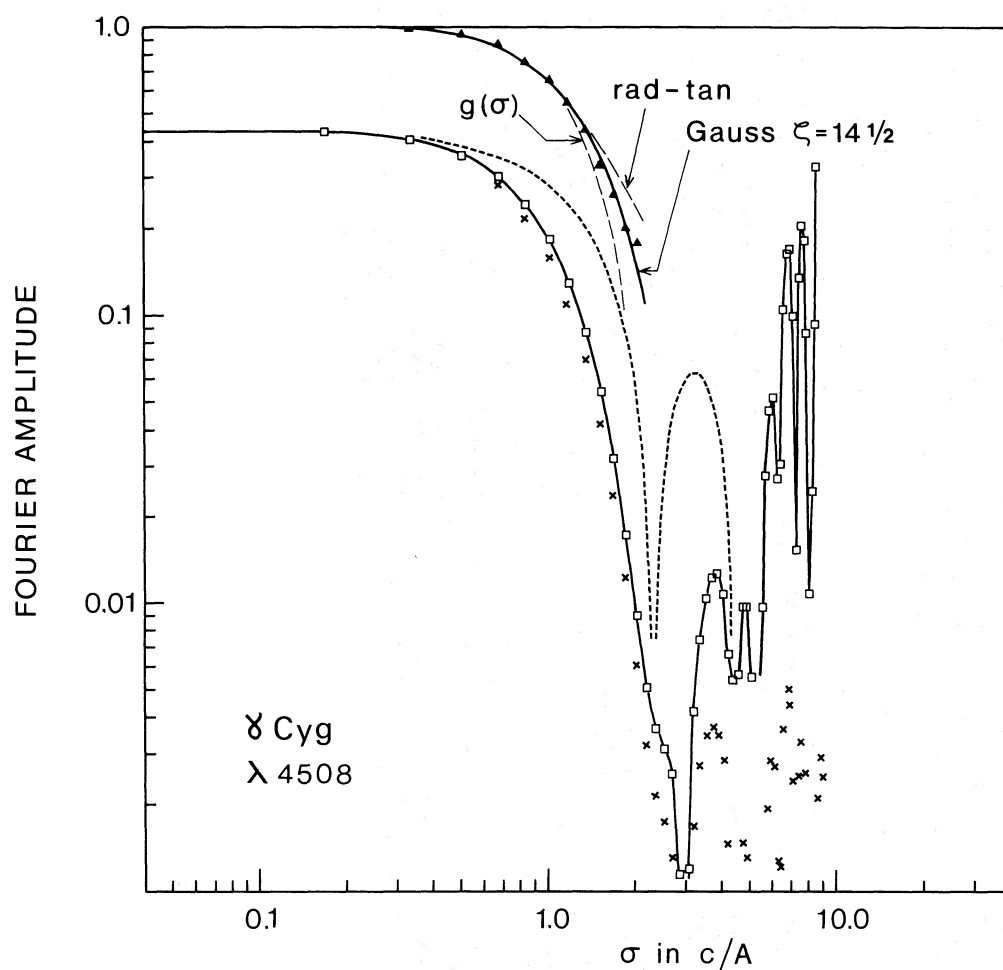


FIG. 10.—The analysis of $\lambda 4508$ in the supergiant γ Cyg proceeds much like those in the previous figures. A $\xi = 5.3 \text{ km s}^{-1}$ and a Gaussian macroturbulence with $\zeta = 14\frac{1}{2} \text{ km s}^{-1}$ are found.

The analysis is depicted in Figure 10. It shows a residual transform which is Gaussian ($\zeta = 14\frac{1}{2} \text{ km s}^{-1}$). The macroturbulence is just large enough that we can choose between the three curves matched to the residuals.

Numerical experimentation verifies that the choice of iron abundance and microturbulence ξ are uniquely defined as they were with the K giants.

The earlier type supergiant, α Cyg, is more difficult to analyze because the macrobroadening is larger. This may seem like an inverted comment, but it results from the combined macroturbulence plus microturbulence reducing the transform amplitude to the noise level before the first zero is reached. Yet the macroturbulence is not so strong that the microturbulence is of small effect. The analysis of Mg II $\lambda 4481$, illustrated in Figure 11, is done with the additional assumption that the Mg abundance must be approximately solar. The result is a microturbulence ξ of $8\frac{1}{2} \text{ km s}^{-1}$ and a radial-tangential macroturbulence with $\zeta = 42 \text{ km s}^{-1}$. If one allows the Mg

abundance to go to ~ 10 – 15 times the solar value, then the solution changes to $\xi = 5.0 \text{ km s}^{-1}$ and a Gaussian macroturbulence with $\zeta = 29 \text{ km s}^{-1}$. So the conclusion for α Cyg remains indefinite until more observations can be brought to bear.

Finally in this set of examples, consider the more complicated case of β Ori A shown in Figure 12. The thermal plus microturbulence filtering is relatively small, making its details uncritical. The residual transform (*triangles*) shows nearly the shape of the rotation profile but not quite. There is a systematic deviation with the data, $d(\sigma)/i(\sigma)$, lying systematically above $g(\sigma)$ as σ approaches $1 \text{ c/\AA}$. This is physically illegal (condition 3) and indicates that the first zero of the data is at a higher σ than it is for $f(\sigma)$. Furthermore, no sidelobe is seen. These inconsistencies, if not attributed to observational errors, can be understood by the modified analysis portrayed in Figure 13. The $v \sin i$ value is decreased slightly, so the first zero in the data and in $g(\sigma)$ match. Now the data points lie satisfactorily below $g(\sigma)$. The residual transform is

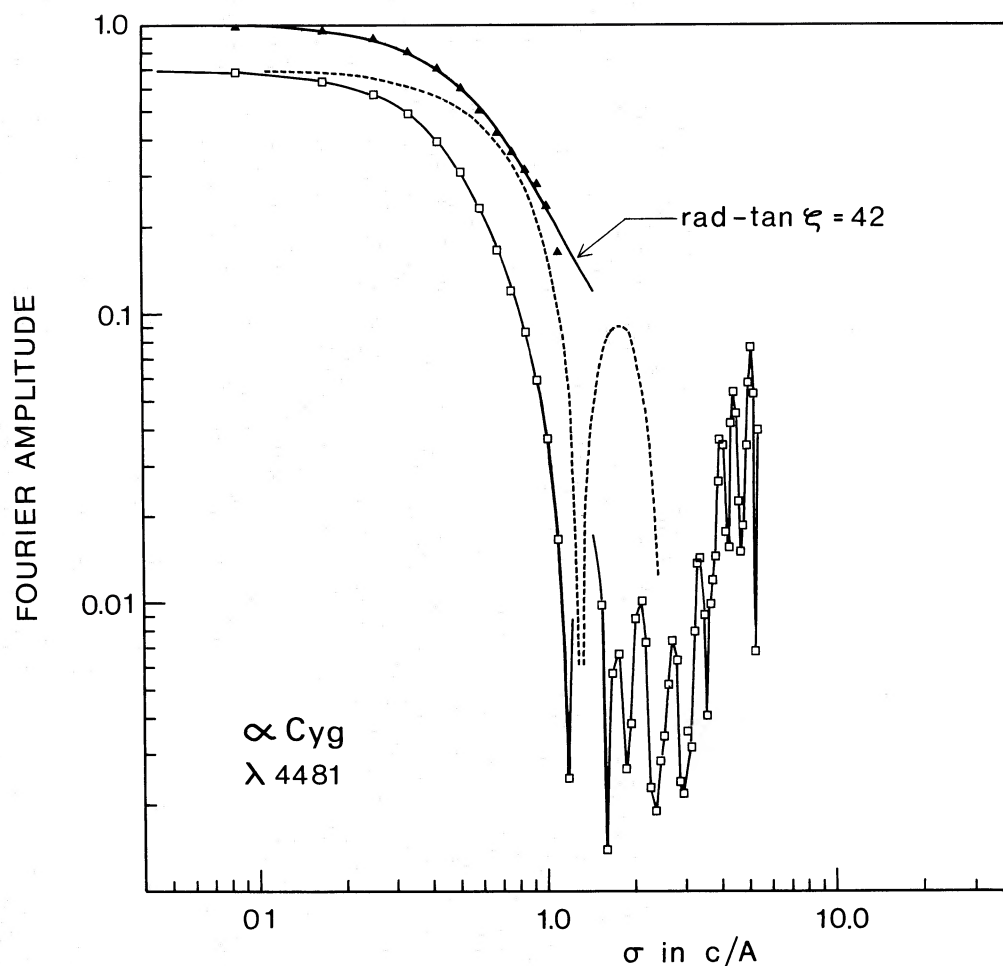


FIG. 11.—This analysis of $\lambda 4481$ in α Cyg results in a radial-tangential macroturbulence with $\zeta = 42 \text{ km s}^{-1}$. A solar abundance for Mg is assumed. As in Fig. 2, the points for $d(\sigma)$ are omitted.

divided by $g(\sigma)$ to obtain a second residual transform (*filled circles*). The second residual is interpreted as macroturbulence; and if it has the radial tangential form, then $\zeta = 23 \text{ km s}^{-1}$, or if isotropic Gaussian, then $\zeta = 17 \text{ km s}^{-1}$. We cannot distinguish between these two macroturbulence solutions.

So in β Ori we have a case where both rotation and macroturbulence appear to be at work. Numerical checks show that neither errors in the limb darkening estimate used in $G(\lambda)$ nor differential rotation (profiles according to Huang 1961) are likely to affect the interpretation in Figure 13. An alternative hypothesis which still needs to be investigated in detail is that β Ori A is a rapidly rotating star seen nearly pole-on. As Stockly (1968) has shown, significant differences exist in the line profiles of stars rotating near the critical velocity, depending on the angle of inclination of the rotation axis. Although one might doubt such an interpretation in this case simply because the star is a supergiant, it may be possible to apply information from additional lines in the spectrum to decide defini-

tively for or against it—namely, the aspect effects depend upon the particular line while the rotation-macroturbulence interpretation does not. One path for future work is therefore clear.

But it may also be that the analysis in Figure 13 has been pushed beyond what the data can support and additional work will be needed to see if the results derived there will stand up. Nevertheless, the data, if taken at face value, lead to the result shown and at least indicate what can potentially be done with the dissection approach to line analysis.

Perhaps it is worth digressing to point out that when half-widths are used to quantify the line broadening and both rotation and macroturbulence are involved, there is no simple and accurate analytical relation to tell us how to separate the two individual half-widths. It is a numerical problem as illustrated in Figure 13. Indeed, unless we know something about $\Theta(\Delta\lambda)$, it is an indeterminate problem. But if one insists on some approximate scheme, consider the three functions: dispersion profile, Gaussian, and rectangular box. The

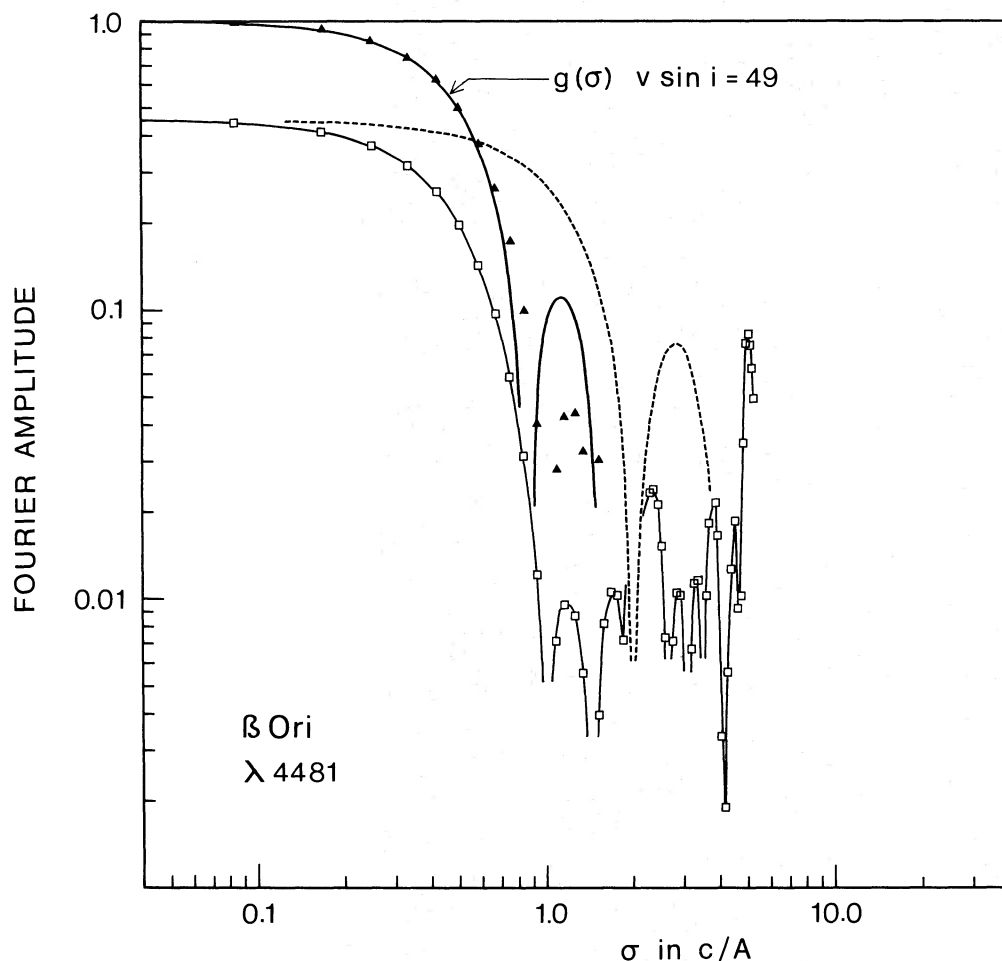


FIG. 12.—The interpretation of the residual transform for $\lambda 4481$ in β Ori A in terms of pure rotational broadening does not work. As in Fig. 2, the points for $d(\sigma)$ are omitted.

first is the most “pointed,” the third the least “pointed.” The convolution of two dispersion profiles results in a new dispersion profile with a width parameter given by the sum of the individual width parameters. The convolution of two Gaussians is a new Gaussian with a width parameter whose square is given by the sum of the squares of the individual width parameters. The convolution of two rectangular boxes is a trapezoid with a half-width equal to the width of the wider of the individual boxes. The rotation profile lies between the Gaussian and the box in shape; and if we assume that the macroturbulence profile is a Gaussian, then the convolution of rotation with macroturbulence gives a function whose characteristic width behavior is between that for two Gaussians and that for two boxes. Linear combinations, in other words, are wrong. Quadratic combinations are wrong too, but not as badly, and invariance with the narrower function, as with boxes, errs on the other side.

Statistical corrections of rotation velocities for the presence of macroturbulence are overestimated if

either the linear or the quadratic formulations are applied.

The last two columns of Table 1 summarize the values of $v \sin i$ and ζ deduced from the examples.

VII. DISCUSSION

The random noise in the original data is probably white. When the transform of the raw data (meaning no averaging of the opposite sides of the profile and prior to removal of the instrumental profile) is computed, the noise is the only component visible in the highest frequencies and it is statistically constant with frequency indicative of whiteness. White noise is found in the higher frequencies for all the data used here, and it is presumed that the whiteness continues to hold to lower frequencies where the signal dominates the transform.

The noise does not stay white. It changes its character at each step in the analysis. When the opposite sides of the observed profiles are averaged

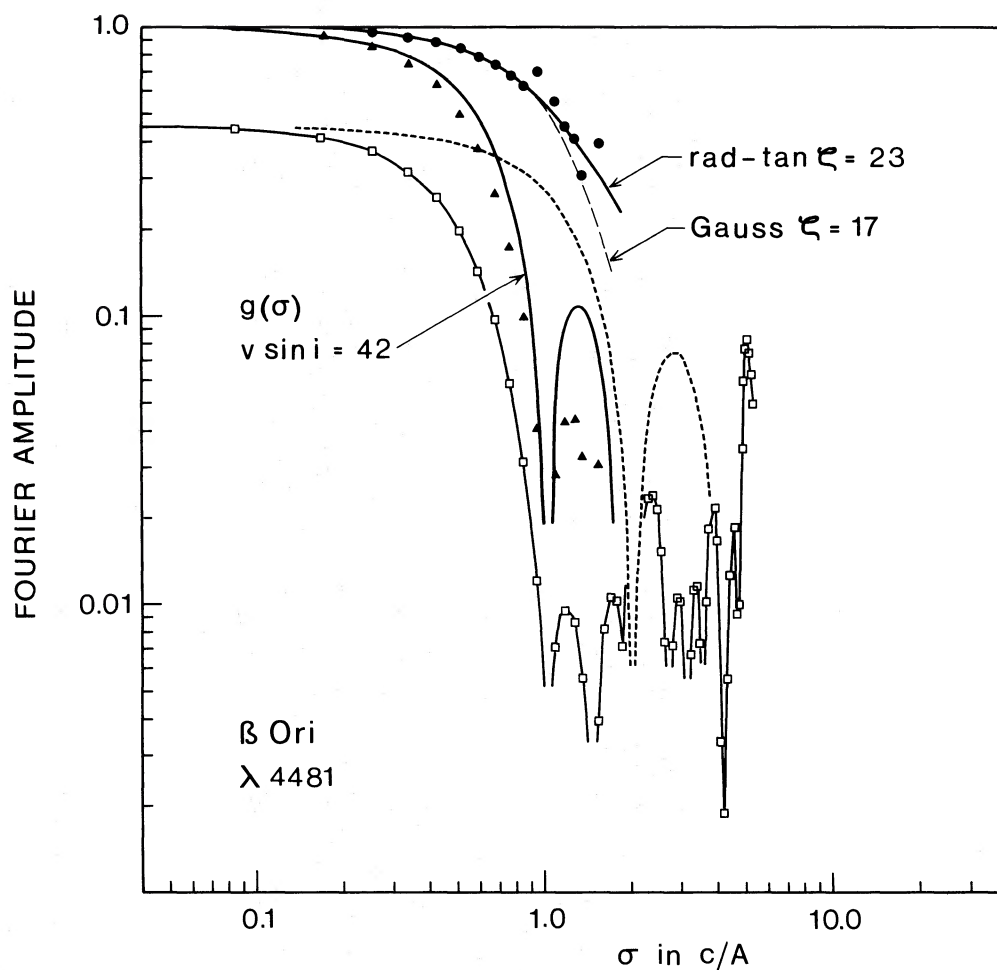


FIG. 13.—This analysis of $\lambda 4481$ in β Ori A leads to a double residual. The first (*triangles*) indicates rotation with $v \sin i = 42 \text{ km s}^{-1}$. The second (*dots*) indicates macroturbulence with $\zeta \sim 17\text{--}23 \text{ km s}^{-1}$ depending on the form of $M(\lambda)$.

for analysis, we expect the noise to be reduced; and this is borne out by comparing the transforms of the raw data and the averaged data. These same comparisons also show two other things. First, the high-frequency noise in the averaged data is reduced by a spuriously large amount, indicating that the process of drawing in the continuous profile has introduced the expected smoothing (§ II) in the high Fourier frequencies as well as accomplishing the desired averaging. Second, the signal (as opposed to the noise) in the transforms $d(\sigma)$ are not significantly or systematically different for the raw and the averaged data. The implication is that the averaging process did not distort the low-frequency Fourier components which were used in the analysis.

The character of the noise is again altered when $i(\sigma)$ and $f(\sigma)$ are removed in the process of obtaining the residual transform. Since the amplitudes of $i(\sigma)$ and $f(\sigma)$ generally diminish with increasing σ , division by them will result in an increase in noise with increasing σ .

In the figures, the effect of noise enhancement by

removal of $i(\sigma)$ is seen in the steep rise of $d(\sigma)/i(\sigma)$ toward the right. The noise becomes ridiculously high. In a conventional analysis, a noise filter must be introduced before transformation back to the λ domain. But in the dissection analysis used here, no such filter is needed. Instead we can use the trend of the noise to estimate the noise level in the signal. To place error bars on each point in the figures would overburden the figures. Instead the residual transforms are shown up to a σ where the noise becomes large. For higher frequency points, the residuals would be essentially pure amplified noise.

Systematic errors can also exist. Continuum placement may introduce a systematic error in the equivalent width (the $\sigma = 0$ amplitude) and could distort the shape of the transform somewhat because the contribution from the wings of the line would be changed. In particular, the flatness of the main lobe of the data transform would be affected. As one can judge from Figure 1, though, the equivalent width would, at most, be increased by a few percent, and this is too small to

perturb the analysis significantly. The conclusions would remain the same with, perhaps, slightly different numbers being derived. In those cases where the spectral resolution is barely adequate, the removal of $i(\sigma)$ can bring in a systematic error if the instrumental profile measurements are inaccurate. But such an error seems unlikely because the examples cover a wide range in the ratio of observed line width to instrumental profile width. No anomalous behavior is seen in the results for the narrow lines compared to those for the wider lines. Scattered light in the spectrograph would not alter the shape of the transform but merely translate the ordinate scale by the amount of the scattered light (see Gray 1974). No scattering corrections were applied to these data.

The valuable sidelobe information which, as pointed out in Paper A, would help identify the different broadening agents in stars do not exist in most of the examples presented here (an exception is α Peg, while β Ori is a borderline case). The other broadening mechanisms have reduced the sidelobes to very low values—well below the noise level. Increased spectral resolution will improve the accuracy because uncertainties in $i(\sigma)$ could be made smaller, but the higher resolution will not be able to bring out sidelobes since they are already gone when the light leaves the star.

Evidence has been presented by some investigators for the hypothesis that the isotropic microturbulent ξ increases outward in the atmospheres of some stars (e.g., Groth 1961, 1972; Parsons 1967; Wolf 1971). It is logical to ask what effect such an alteration would have on the material presented here. The answer is, very little. The line profiles from such models are more boxlike than for a constant ξ . But the saturated lines were already deduced to be boxlike. Most of the increased boxiness will be manifest in the sidelobe structure, which, as we have seen, is lost from the analysis. An isotropic microturbulence increasing inward will not work, but an anisotropic microturbulence increasing inward probably could be made to work. There is no need in this paper nor justification in the data analyzed here to go to such elaborate model manipulation.

It is clear that the characteristics deduced for the macromotions depend on the shape of the thermal plus microturbulence profile, at least for stars like those in the examples. And in fact we have found that this profile must be more or less boxlike for the saturated lines. Such behavior can be produced with the hypothesis of classical microturbulence as we have done here. But should it subsequently be found that a different physical mechanism produces the boxiness, the rest of the dissection analysis remains intact. In other words, it is the boxiness of the profile that has been used and not the physical mechanism producing

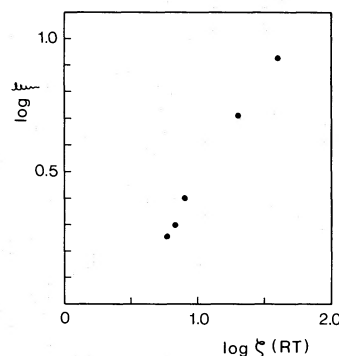


FIG. 14.—There is a correlation between the microturbulence ξ and the macroturbulence ζ . The macroturbulence ζ used in this figure is the value that one would deduce if all of the $M(\lambda)$'s were of the radial-tangential form.

the boxiness. As shown in Figure 14, there is a good correlation between the ξ and ζ values. Such a relation might be expected between small- and large-scale turbulence, but it might also be expected from some alternative hypotheses.

The damping constants used in the model calculations incorporated depth-dependent van der Waals and Stark terms and a radiation damping of 10 times the classical value. But because the data do not show strong wings, all of the model profiles used in the analyses do not show strong wings and the effects of damping or its uncertainty have almost no bearing on the results.

No claim is made to phenomenal accuracy for the $v \sin i$, ξ , and ζ values presented in the examples. They are probably good to a few percent except for the macroturbulence in β Ori which is less certain. (These error estimates are based on the change in the solution as a function of trial alterations in the observed profile and on the internal consistency of the $\lambda 6064$ versus $\lambda 6065$ solutions for the same star.) The Fourier method may well be capable of supplying better leverage on precision than the classical profile fitting, but that remains a topic for a future investigation.

The examples do show how the Fourier dissection of broadened line profiles allows the separation of rotation from macroturbulence and how some of the features of the macroturbulence velocity distribution can be measured. These results indicate the possibility of more than one kind of macrovelocity distribution. Additional measurements and verification are needed before one should speculate on the implications of such a possibility.

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